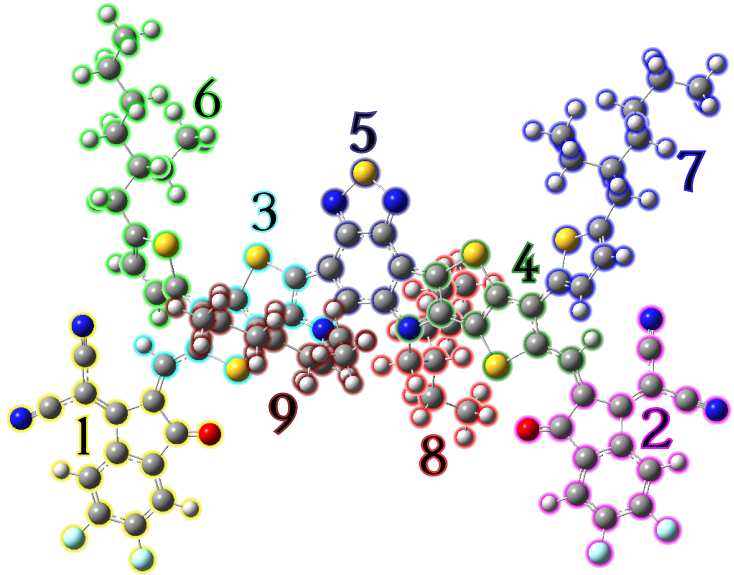
a)



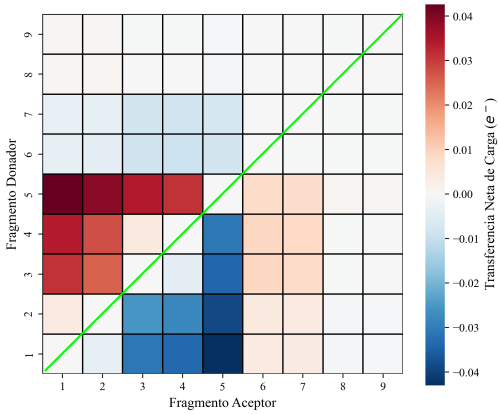
b)

HOMO→LUMO 100%

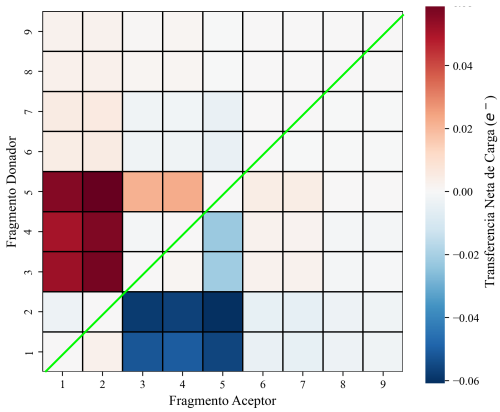
HOMO→LUMO 74%
HOMO-1→LUMO+1 24%

HOMO→LUMO 95%

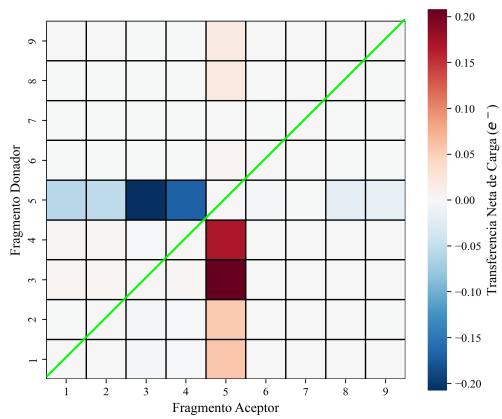
1 EE



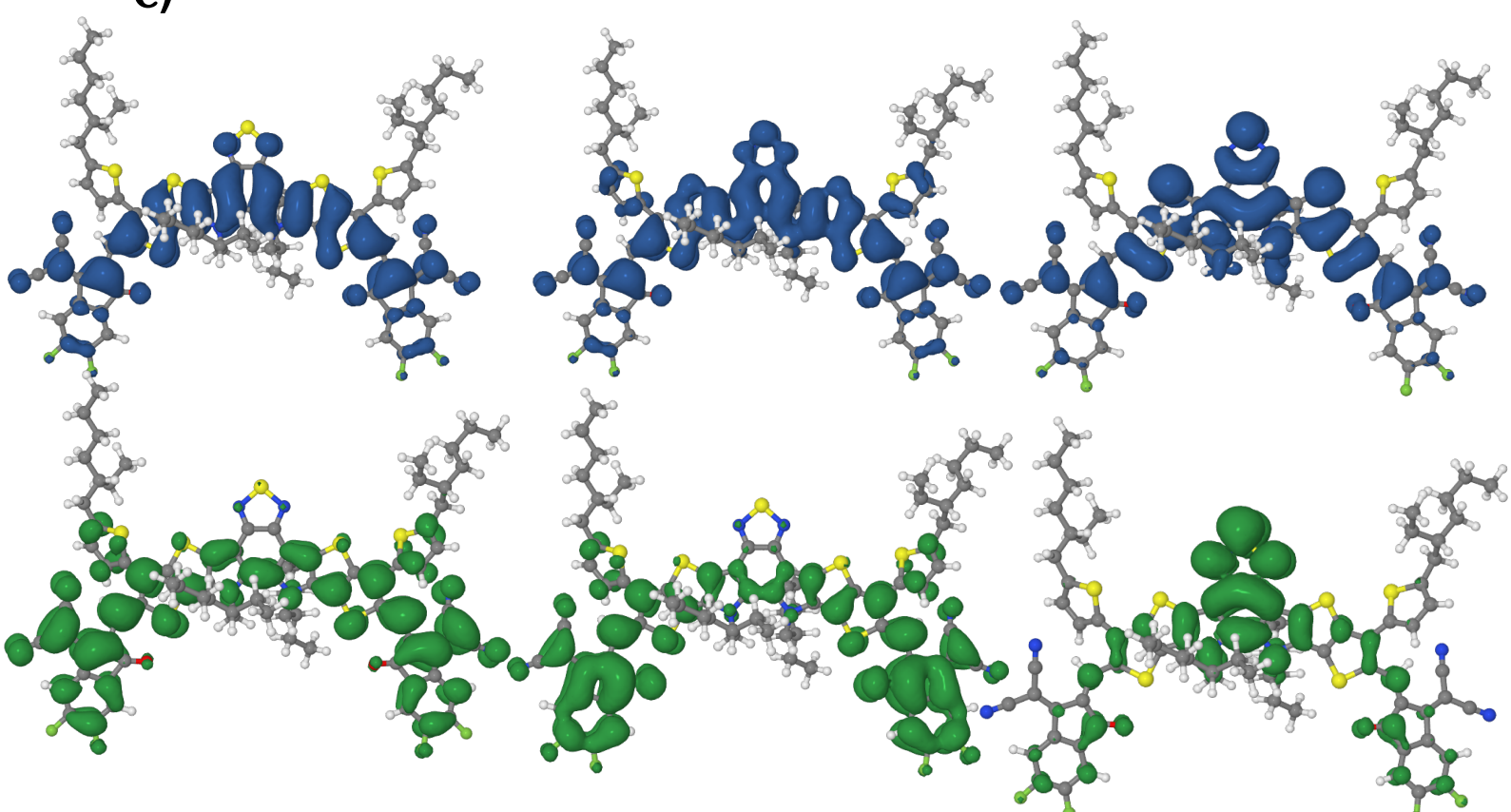
8 EE



38 EE



c)



$D_{index} = 1.231$ Å

$D_{index} = 2.454$ Å

$D_{index} = 2.438$ Å